

Search & discovery — three new ways to find a part v0.99.25/28/29

Beyond keyword search: by MEANING, by PHOTO, and by RELATIONSHIP. All offline.

📄 Semantic search /semantic

embed.py + /api/semantic. Search by meaning:
'coolant leak at the water pump' finds the right pages w/o those exact words.
Local sentence-transformers model when installed; deterministic hash fallback else.
Build: BUILD-EMBEDDINGS.bat. Verified: self-cos=1.0, related 0.73 > unrelated 0.0.

📄 Visual match /visual

phash.py + POST /api/visualmatch. Drop a photo of a part → closest figure crops by 64-bit DCT perceptual hash + Hamming.
Pure numpy/PIL, offline, no model.
Build: BUILD-VISUAL-INDEX.bat -> phash.tsv.
Verified: identical/resized = 0, unrelated = 31.
/figcrop serves the matched crop.

📄 Related /related

xref.py + /api/xref. For a part: the assemblies/figures it belongs to, its SIBLINGS (same figure), and SEE-ALSO (same assembly elsewhere) — each links to dossier / deep-zoom.
Verified: siblings BOLT/NUT, see-also VOLTAGE REGULATOR (self-inclusion bug fixed).

VERIFIED / R1

- All three cores verified in-sandbox against real logic (cosine / Hamming / xref on synthetic DBs); pages pass node --check.
- Model/index BUILD steps run host-side (BUILD-EMBEDDINGS / BUILD-VISUAL-INDEX) — the app degrades gracefully with a clear 'not built' state.
- Wired: /semantic /visual /related in _PAGES + palette + Tools menu + verify_ui; /api/semantic /api/visualmatch /api/xref /figcrop.
- Additive & read-only: new modules + routes + pages. Nothing writes the index (R1/R6).